

OPAN 3257

Develop/Managing Data Bases

Databases for Business: From First Query to Live Dashboard

Fall 2026

In Person, **Hariri 230**, McDonough School of Business, Georgetown University

Course Name & Number	OPAN 3257 – Develop/Managing Data Bases
Class Schedule & Room	Monday & Wednesday, 3:30 PM – 4:45 PM — Hariri 230
Semester	Fall 2026
Start & End Dates	Wednesday 08/26/2026 (first class day) Monday 12/07/2026 (last class day) Friday 12/11/2026 (last deliverable due — final project resubmission)
Grading / Credits	Graded / 3 credits

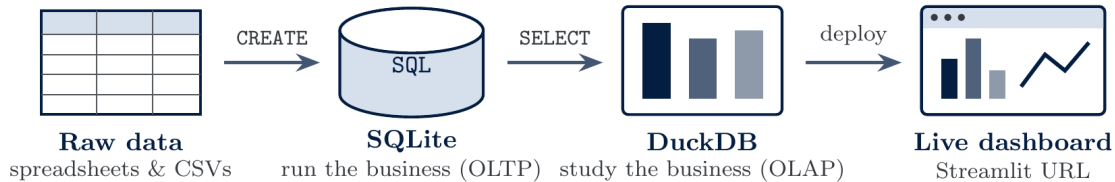
Overview: Databases Run the Business

Behind every consequential business application sits a database: the point-of-sale terminal ringing up your latte, the cart holding your online order, the reservation system that just rebooked your flight, the ledger that knows your bank balance to the cent, the CRM logging every sales call, and the routing engine deciding which van carries tonight’s packages. Computer-science departments treat this as a truism—their flagship database courses are built on the premise that a database management system sits at the core of nearly every important application. This course translates that premise into the language of business: *if the database is where the business actually lives, then reading, questioning, and designing databases is a core management skill, not a specialist’s trick.*

Consider what a modern firm is, informationally. Today, some of the most valuable corporations are valuable not because of their physical assets, but because of the information they own: imagine a bank without its data on accounts and customers, a retailer without its order history, an airline without its reservations, or a social-network site that loses the connections among its users. Marketing runs on customer tables; operations runs on inventory, shipment, and scheduling tables; finance runs on ledgers—which are tables with very strong opinions about correctness. Since *Harvard Business Review* dubbed data science “The Sexiest Job of the 21st Century” in 2012, nearly every industry has learned the same lesson: the analytics, the dashboards, and now the machine-learning and AI systems all begin with well-managed data. Database management has evolved from a specialized computer application into a central component of virtually every enterprise—and an essential part of an education in business.

Most businesses first meet their data in spreadsheets, and spreadsheets are where businesses first break: two analysts edit two diverging copies; a lookup silently misses; ten million rows of transactions refuse to open; an intern deletes a column and no one notices until the audit. A **database-management system (DBMS)**—a collection of interrelated data and a set of programs to access those data—exists to solve exactly these problems: to make storing and retrieving information *convenient* and *efficient*, for many users at once, while ensuring the *safety* of the information stored.

Data does two different jobs inside a firm, and this course teaches both. *Transactional* databases **run** the business: they record orders, payments, reservations, and account balances, where every write must be correct, immediate, and safe. *Analytical* databases **study** the business: they answer the Monday-morning questions—Which products, regions, and customers drive revenue? Where do fulfillment promises slip, and what do late deliveries cost us in review stars? Which appointment slots go unused? Practitioners call this divide OLTP¹ versus OLAP², and you will work on both sides of it: **SQLite** for building and operating a business database, and **DuckDB**—a free, modern, warehouse-grade analytical engine that runs inside our notebooks—for interrogating millions of rows the way analysts do at work.



From raw data to live dashboard—the arc of this course.

The semester follows an arc of seven verbs, each recast for business: *query* (ask precise questions of data), *relate* (connect customers to orders to products), *design* (model a business process as tables, keys, and constraints), *write* (insert, update, and delete—and why transactions keep a ledger honest), *view* (package queries into reusable reports), *optimize* (why indexes make queries fast, and what that costs), and *scale* (from a file on your laptop to engines like PostgreSQL and MySQL and the cloud warehouses built on the same SQL you will learn here). Formally, the course centers on the SQL programming language, the critical tool for managing data in Relational Database Management Systems (RDBMS), across its three major sub-languages:

1. **Data Definition Language (DDL):** for creating and modifying database structures.
2. **Data Manipulation Language (DML):** for performing operations like reading, inserting, updating, and deleting data.
3. **Data Control Language (DCL):** for managing access to data.

We present the material in a clear, simple style, with an extensive set of exercises (solutions on Canvas) accompanying each topic. To facilitate hands-on learning, this course utilizes **Google Colab**, a free, cloud-based platform that allows you to write and execute SQL code in real time, alongside lectures. SQL has evolved from a specialized tool for computer experts into a widely used standard across industries, and the knowledge you gain here will be applicable immediately in real-world scenarios.

This course assumes you know NOTHING about databases or coding. We start from zero and build your skills step-by-step; every activity is designed to reinforce what you just learned. By the end, you'll be amazed at what you can do—and you will leave with an artifact you built and can keep: a functioning multi-table business database, a portfolio-ready analysis defended orally, and a live **Streamlit** dashboard—at a URL you can share—on a real business-operations problem.

¹Online Transaction Processing: the operational workload of recording the business as it happens—orders, payments, updates—one fast, correct transaction at a time.

²Online Analytical Processing: the analytical workload of asking questions across history—aggregating millions of rows to spot trends, segments, and problems.

Course Objectives

At the end of this course, you will be able to:

1. **Understand SQL Fundamentals:** gain a thorough understanding of Structured Query Language, including its key sub-languages (DDL, DML, DCL) and their applications in database management.
2. **Develop Practical SQL Skills:** write, execute, and debug SQL code to create, read, update, and delete data within relational databases using Google Colab.
3. **Apply Database Concepts to Real Business Scenarios:** use SQL to solve real data-management problems drawn from retail and e-commerce, last-mile delivery and logistics, marketing, finance, hospitality, and sustainability—and end every analysis in **actionable recommendations**, not toy output.
4. **Model a Business Process as Data:** translate an operating business—its customers, orders, inventory, and promises—into a sound relational design with entities, keys, and constraints.
5. **Explore Advanced SQL Features:** master joins, subqueries, views/CTEs, and CASE segmentation, and apply them to complex retrieval tasks, including **warehouse-grade analytical SQL with DuckDB**.
6. **Explain How Databases Power the Enterprise:** distinguish the databases that run the business (OLTP, transactions, integrity) from the databases that study it (OLAP, warehouses, columnar engines), and understand just enough of what is under the hood—indexes, query plans—to work credibly with engineering and IT teams.
7. **Utilize Google Colab for Collaborative Coding:** leverage Colab for writing and sharing SQL code, enabling collaborative learning and live demonstrations.
8. **Ship a Deliverable:** publish your final analysis as a small **live Streamlit dashboard** at a public URL.

Course Structure

This course is laid out into **weekly modules** on Canvas, with experiential/team learning activities built in to enrich key topics. See the **Course Calendar** for the at-a-glance schedule and the **Session-by-Session Guide** for exactly what we will do in each meeting. Each week also lists an optional SQLBolt warm-up lesson and an optional CS50 SQL deep-dive video, so you always have a second explanation of every concept, on demand and free.

Time Expectation (3-Credit Course)

As with all Georgetown 3-credit courses, students are expected to spend around **10 hours per week** on all learning activities, including class time, readings, pre-class notebooks, assignments, and the project. The preparation instructions on Canvas are designed to help you master the topics efficiently. **If you don't follow them, you may either learn less or spend more time achieving the same results.**

Learning Technology Toolkit

Everything you need in this course is deliberately **no-cost to students**—Google Colab is free, the datasets are open, and no paid tools are required. This semester the toolkit runs almost entirely on your Georgetown Google account: one login powers Colab, your AI tutor, and your study materials.

- **24/7 AI Course Tutor (a custom Gemini “Gem”) [free]**— I have built a course-customized Gemini assistant—the *OPAN 3257 SQL Tutor*—loaded with our syllabus, weekly topics, datasets, and my hints-first tutoring style. It explains SQL concepts, generates unlimited practice queries at your level, and helps you get unstuck at 2 a.m., not just during office hours. Georgetown students can activate Google’s **AI Pro student plan at no cost**, which unlocks the strongest Gemini models for the Gem, Colab, and NotebookLM alike; the activation link and the Gem link are on Canvas. Use it to *learn*; the Generative AI policy below governs what may be submitted, and **it is not permitted on the Semester Project**.
- **Google Colab [free]** — our primary coding environment: write and execute SQL directly in your browser, no installation, from any device; built-in SQLite support; easy sharing; automatic saving to Google Drive. **All lecture materials are interactive Colab notebooks** you can annotate, collaborate in, and keep after the course ends.
- **Colab’s built-in Gemini assistant [free]** — a second helper inside the coding environment itself, for troubleshooting errors and hints while you code.
- **SQLite and DuckDB [free]** — we start with SQLite (a single-file database perfect for learning and for *running* a business database) and, in the second half of the semester, step up to **DuckDB**, a free, open-source analytical engine that runs inside our Colab notebooks with zero installation—so you finish the course writing **warehouse-grade SQL** directly over CSV and Parquet files, the way analysts do on the modern data stack.
- **SQLBolt [free]** (<https://sqlbolt.com>) — short, interactive browser lessons assigned as optional pre-class warm-ups; each week’s module lists the matching lesson so you can drill the syntax before we apply it to business data.
- **Week 0 module [free]** — practice materials and recorded walk-through videos so students with no programming background can close the gap privately, at their own pace; a parallel **Enrichment module** gives already-fluent students somewhere further to go. *The standard never moves; the number of routes to it multiplies.*
- **Mentimeter & Kahoot [free]** — rapid, low-stakes comprehension checks in every session (scan a barcode to join), with **small prizes**; several classes close with a short **exit-ticket** quiz.
- **Streamlit Community Cloud + GitHub [free]** — **required for the Final Project**: you will deploy a small dashboard of your results to a live public URL using a provided starter template (both accounts are free; step-by-step guide on Canvas; no web-development experience needed).
- **FeedbackFruits [free]** (Georgetown-licensed) — structured, anonymous **peer review and peer evaluation** for team activities and the semester project.
- **NotebookLM [free]** — podcast-style audio overviews and visual summaries of selected readings and cases, posted before class as multi-modal preparation—generated from the same course materials, on the same Google account.

- **“Ask your data” with the free-tier LLM/Gemini API [free]** (*optional extra credit*) — advanced teams may add a natural-language query cell to their dashboard; guide in the Enrichment module.
- **Vocareum** (*possible pilot; no cost to students*) — auto-graded practice notebooks for instant, specific feedback on selected exercises may be piloted; any pilot will be announced on Canvas.
- **Canvas and Zoom [free]** — Canvas (<http://canvas.georgetown.edu/>) is the primary platform for due dates, instructions, resources, assignments, grades, discussions, and Zoom links; Zoom hosts office hours and recorded materials and is our backup for live sessions.

Real Data, Real Problems: Top Data Sources (Curated)

Rather than a long menu, we focus on a **short list of the best open sources**—each chosen because undergraduate students learn something **different** by applying it. Assignments, mini-challenges, and the semester project draw on these:

Source	Domain	What you learn by applying it
Olist Brazilian E-Commerce (Kaggle) https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce	Retail & e-commerce	~100,000 real orders across eight linked tables (orders, items, customers, sellers, products, payments, reviews, geolocation)—the best public practice ground for keys, JOINS, and the full order lifecycle; backbone of project Track R
Inside Airbnb https://insideairbnb.com/get-the-data/	Marketplaces, hospitality, pricing	Genuinely relational multi-table data (listings–calendar–reviews): occupancy, pricing, and review analytics; backbone of project Track H
NYC TLC Trip Record Data https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page	Mobility & last-mile operations	Tens of millions of trips in raw CSV/Parquet—where DuckDB shows what warehouse-grade speed feels like; feeds the Data Import Bootcamp
World Bank Open Data https://data.worldbank.org	Economics, finance, sustainability	Country–year panel data; GROUP BY over time; joining across indicator tables; the ESG angle
Yahoo! Finance https://finance.yahoo.com	Finance & markets	Time-series data; date handling; monthly/quarterly aggregation; rankings with ORDER BY/LIMIT
UC Irvine ML Repository https://archive.ics.uci.edu	Classic datasets	Clean, documented schemas for focused practice with functions and CASE segmentation

Kaggle more broadly remains our first stop for finding and judging business datasets. **Want a different dataset?** Search [Google Dataset Search](#) and propose it—instructor approval required. For presenting results, [Flourish](#) (free tier) or simple charts in Colab are both welcome.

Semester Project: The Data-Driven Enterprise — Pick Your Industry, Run Its Data

Every operating business is a queueing system wearing a brand: orders arrive, resources serve them, bottlenecks form, promises are kept or broken. The questions that decide customer experience—*How long do customers wait? Which SKUs stock out? Which routes run hot? Which promises slip, and what do they cost in review stars?*—are exactly the questions a well-designed database can answer. **The official project brief, dataset links, starter templates, and team sign-up are posted on Canvas.**

Working in **teams of 3–4**, you act as the analytics consulting group reporting to a COO: you **design and build** a relational database for a real operating business, **query it** to locate operational bottlenecks and **recommend actions**—and finally **ship your results as a live dashboard**. It is **one project divided into two parts**. Each team chooses **one of three industry tracks** (or proposes its own):

- **Track R — Retail & E-Commerce Operations.** Built on the Olist Brazilian e-commerce dataset (~100k real orders, eight relational tables). Sample questions: Where do delivery promises slip, and which seller–region pairs drive lateness? Do late deliveries cost review stars? Which categories carry the margin? What does the payment/installment mix say about customers?
- **Track D — Last-Mile Delivery & Logistics.** Built on a starter relational dataset provided on Canvas (depots, drivers, vehicles, routes, packages, delivery attempts, customers, time windows), inspired by my three years at Convoy, the digital freight marketplace. Sample questions: What is our failed-first-attempt rate, and what does redelivery cost? Which routes and drivers run hottest? Where would a micro-depot cut promised-vs-actual gaps?
- **Track H — Hospitality & Marketplace Operations.** Built on an Inside Airbnb city extract (listings–calendar–reviews). Sample questions: Which neighborhoods are supply-constrained? How do price calendars respond to demand? What separates high-occupancy hosts? What does review velocity predict?
- **Propose your own** (with instructor approval): any operating business with genuinely relational data—including healthcare operations, if that is where your interests run.

Part 1 — Midterm Project: Design & Build (20% of course grade)

Due: Monday, October 26, 2026, 3:30 PM (Canvas). Deliverable: one Google Colab `.ipynb` notebook containing all of the following:

1. **Team charter & data memo:** team name, members, chosen track and dataset(s), and the 3–5 operational questions your database will answer.
2. **Database design:** an ER diagram (entities, relationships, primary/foreign keys) with a short justification of your design choices—your tables should read as a model of the business process itself.
3. **DDL:** CREATE TABLE statements implementing the design, with keys and constraints.
4. **Population:** data loaded via INSERT statements and/or CSV import.
5. **Baseline queries:** at least 8 single-table queries (SELECT, WHERE with AND/OR, text functions, arithmetic, NULL handling)—each preceded by the business question it answers, in plain English.

Attribution: label each code cell with the name of the student who developed it. Written feedback is returned within one week so Part 2 builds on a corrected foundation.

Part 1 grading rubric	Weight
Database design (ER diagram + DDL with keys and constraints)	40%
Data loading (INSERT / CSV import, correct and complete)	15%
Baseline queries (8+, each tied to a stated business question)	35%
Notebook quality, documentation & per-cell attribution	10%

Part 2 — Final Project: Analyze, Recommend & Deploy (30% of course grade)

Extend your Part 1 database into a full operational analysis. **Required deliverables (all three):**

1. **The notebook** (Colab .ipynb only — **no PowerPoint**):
 - **Multi-table analysis:** joins (including at least one **three-table join**), aggregates with GROUP BY/HAVING, at least one **subquery**, at least one **CASE segmentation**, and ORDER BY/LIMIT rankings.
 - **At least one analytical query executed in DuckDB** directly over the raw CSV/Parquet files—your taste of the warehouse.
 - **Each team member must create at least one intermediate-to-advanced SQL query** (e.g., a two-person team submits at least two such queries), **clearly labeled with that student’s name**.
 - A closing **Actionable Recommendations** section addressed to the COO and leadership team, grounded in your query results.
2. **The live dashboard (required):** a small **Streamlit** app deployed to **Streamlit Community Cloud** showing **at least three key operational metrics/charts** from your analysis plus a short recommendations block. Submit the **live public URL** with your notebook. One deployment per team.
3. **The oral defense (individual):** in class on the assessment days, each member answers questions about the team’s queries.

How to deploy (step by step; budget 2–3 hours with the template)

1. Create free accounts on [GitHub](#) and [Streamlit Community Cloud](#) (both free; no credit card).
2. Copy the **starter template repository** (link on Canvas) into your team’s GitHub account.
3. Add your data: export key query results from Colab as CSV, **or** run your DuckDB queries directly in the app—both patterns are already wired into the template.
4. Fill in the template with your three (or more) metrics/charts and your recommendations text.
5. Click **Deploy** on Streamlit Community Cloud and paste the live URL into your Canvas submission.

Support: a step-by-step video and written guide on Canvas; a **live deployment help station** at the Project Workshop Day (Wed, Nov 18); TA office hours. No web-development experience is needed or expected.

Optional extra credit (up to +3 points on the Part 2 score): (a) an “ask your data” natural-language query cell using the free-tier LLM/Gemini API (Enrichment-module guide), and/or (b) additional dashboard interactivity such as filters or drill-downs.

Part 2 grading rubric	Weight
SQL correctness & sophistication (joins, aggregation, subquery, CASE, DuckDB query; one attributed intermediate-to-advanced query per member)	40%
Operational insight & actionable recommendations	25%
Streamlit dashboard (live URL, required content, works when opened)	15%
Individual oral defense	15%
Notebook quality, documentation & per-cell attribution	5%

Project Timeline (all items on Canvas)

Date	Milestone
Mon, Sep 21 (Meeting 7)	Project kickoff: Data-Driven Enterprise brief and track menus released; team sign-up opens
Fri, Sep 25, 11:59 PM	Team sign-up due (teams of 3–4), with tentative track choice
Mon, Oct 26, 3:30 PM	Part 1 (Midterm Project) due — Colab notebook
Wed, Nov 18 (Meeting 23)	Project Workshop Day — milestone check + live deployment help
Mon, Nov 30, 3:30 PM	Part 2 initial submission — notebook + live Streamlit URL
Wed, Dec 2 & Mon, Dec 7	In-class walkthroughs & individual oral assessments (by group number); awards on Dec 7
Wed, Dec 9, 11:59 PM	Peer evaluation on FeedbackFruits due
Fri, Dec 11, 11:59 PM	Final resubmission (notebook + URL refinements) after incorporating feedback

Project Rules

The project is **Type A** (one submission per team) with individually attributed queries and an individual oral defense (**Type D**). **Generative AI tools are NOT allowed on the Semester Project (Parts 1 and 2)**; tools like Grammarly for language aid are permitted. Credit is earned through evidence that cannot be delegated: your own queries, defended orally in front of the class. Peer evaluations adjust individual scores on team components. Using un-cited sources constitutes academic dishonesty.

Grading Policies

Component	Weight
Participation & in-class activities (Mentimeter/Kahoot, exit tickets, attendance)	10%
Mini-Challenges (6 assigned; lowest score dropped)	15%
Assignments 1–5	25%
Semester Project, Part 1: Midterm Project (Design & Build)	20%
Semester Project, Part 2: Final Project (Analyze, Recommend & Deploy)	30%
Total	100%

There is no final exam. Please read the Grading Policies page on Canvas thoroughly, as knowing every detail (including letter-grade conversion) is crucial for understanding how your final grades will be determined.

Assignments & Mini-Challenges: What You Will Actually Build

The five assignments carry the business arc of the course (with attribution in each handout):

- **Assignment 1 — The Catalog & the Customer (due Mon, Sep 28).** Single-table analysis of a retail product catalog and a CRM export: `SELECT`, `WHERE`, text functions, arithmetic. Companion drills: SQLBolt lessons 1–6.
- **Assignment 2 — The Monday-Morning Report (due Mon, Nov 9).** The recurring KPI questions every operator asks—revenue by region, top products, repeat-customer share—answered with `GROUP BY`, `HAVING`, aggregates, and `ORDER BY/LIMIT` on an e-commerce dataset. Doubles as training for your Part 1 baseline queries.
- **Assignment 3 — Moneyball (due Mon, Nov 16).** The original data-driven hiring problem: join player, salary, and performance tables to find undervalued talent—then translate the same logic to hiring and vendor selection in any firm. Lands one week after three-table `JOINS`.
- **Assignment 4 — Packages, Please (due Mon, Nov 23).** Trace lost parcels through a last-mile delivery database with joins and subqueries—detective work that doubles as a warm-up for Track D teams heading into the final project sprint.
- **Assignment 5 — The SQL Gauntlet (due Fri, Dec 4, 11:59 PM).** Ten progressively harder analytical queries over a large multi-table dataset, **run in DuckDB**; released with a DuckDB primer notebook so everything you need ships with the assignment. Your capstone in place of a final exam.

The six **Mini-Challenges** are short, individual, in-class problems (**lowest dropped**): design a campus coffee-shop menu table; extract fields from a customer table; build fraud- and returns-screening filters; build and populate an inventory table; a store analysis with `JOIN`; and a customer-order analysis with `GROUP BY`.

Assignment Types and Collaboration Rules

Every graded item belongs to one of the following types, which define exactly who you may work with:

Type	Designation	Collaboration	Submission
A	Team submission	Discussion is permitted with your designated team but not with other students	One submission per team
B	Individual submission with team discussion	Discussion is permitted with your designated team but not with other students	Individual submissions; the team's work can be a basis for individual submissions
C	Individual submission, no discussion	Students may not discuss the content of the assignment with other students; broad concepts may be discussed	Individual submissions
D	Quiz, Test, Exam, Oral Assessment	Students may not discuss the content or broad concepts with other students; no assistance of any kind	Individual

In this course: Assignments 1–5 = Type B; Mini-Challenges = Type C; Semester Project Parts 1–2 = Type A (with individually attributed queries); Oral assessments = Type D.

Course Calendar (At a Glance)

(Tentative: check Canvas for the most up-to-date information and deadlines; confirm holidays against the official Georgetown Fall 2026 academic calendar.) Full explanations of every meeting follow in the **Session-by-Session Guide**. MC = Mini-Challenge; A = Assignment.

Wk	Date (Mtg)	Topic	In-Class Highlights	Prep & Deliverables
1	W 8/26 (M1)	What is Data? Business Runs on It	“ Data in Your Pocket ”: ten things a business knows about you; first look at Colab	Code of Conduct + intake survey ; Week 0 if new to coding
2	M 8/31 (M2)	From Spreadsheets to Databases	“ Human Spreadsheet ”; where spreadsheets break a business	Topic overview; SQLBolt L1 (opt.)
2	W 9/2 (M3)	Your First Table (CREATE TABLE)	Café menu → SQLite; MC1 (coffee-shop menu)	Extension notebooks 1–2
3	M 9/7	<i>No Class — Labor Day</i>		
3	T 9/8 (M4) <i>(Mon. classes meet)</i>	Data Types & Primary Keys	“ Type Casting Call ”: match business facts (price, SKU, date, email) to INTEGER/TEXT/REAL; give the café menu a proper PRIMARY KEY	Topic overview

Wk	Date (Mtg)	Topic	In-Class Highlights	Prep & Deliverables
3	W 9/9 (M5)	SELECT: First Real SQL	“ Data Treasure Hunt ” in a product catalog; everyone runs a query	Topic overview
4	M 9/14 (M6)	Choosing Columns	“ Column Detective ” on a CRM export	String-function sheet
4	W 9/16 (M7)	WHERE	“ Filter Game ”: segmenting customers; common mistakes; MC2 (customer extract)	Topic overview
5	M 9/21 (M8)	Multiple Conditions (AND/OR)	“ Complex Filters ”: fraud & returns screens	Project kickoff — teams due Fri 9/25
5	W 9/23 (M9)	Text Functions	“ Text Transformer ”: clean names & shipping addresses	Topic overview
6	M 9/28 (M10)	Database Puzzle Challenge	5 stations, team rotation	A1 due 3:30 PM
6	W 9/30 (M11)	Math in SQL	“ Calculate the Bill ”: margins, discounts, taxes; MC3 (fraud & returns filters)	Topic overview
7	M 10/5 (M12)	NULL & Missing Data	COALESCE/IFNULL; Case Study 1: retail inventory	Topic overview
7	W 10/7 (M13)	Views, CTEs & Temporary Tables	The analyst’s report shelf; partitioning demo	Topic overview
8	M 10/12	<i>No Class — Columbus Day</i>		
8	W 10/14 (M14)	INSERT	“ Populate the Database ”: receiving day at the warehouse	Topic overview
9	M 10/19 (M15)	UPDATE, DELETE & Transactions	SQL Detective Mystery; why a ledger never half-updates (ACID); MC4 (inventory table)	Topic overview
9	W 10/21 (M16)	Your First JOIN	Physical join demo; INNER vs LEFT	Topic overview
10	M 10/26 (M17)	Types of JOINS	“ Missing Data Detective ”	Midterm Project (Part 1) due 3:30 PM
10	W 10/28 (M18)	GROUP BY + HAVING + Guest Speaker (Databricks)	“ Grouping Olympics ”	Topic overview
11	M 11/2 (M19)	Sorting & Limiting (TA leads)	“ Top 10 Challenge ”: best-seller leaderboards; MC5 (two-store JOIN)	Topic overview
11	W 11/4 (M20)	Aggregate Functions (TA leads)	“ Store Statistics ”: the five KPI verbs	Topic overview
12	M 11/9 (M21)	GROUP BY in Depth	“ Group and Count ”: cohorts & segments	A2 due 3:30 PM

Wk	Date (Mtg)	Topic	In-Class Highlights	Prep & Deliverables
12	W 11/11 (M22)	Three-Table JOINS	“ Data Web ”: customer → order → product; MC6 (orders per customer)	Topic overview
13	M 11/16 (M23)	Subqueries	“ Nested Questions ”: beat-the-average benchmarks	A3 due 3:30 PM
13	W 11/18 (M24)	Project Workshop Day	Team work w/ instructor & TA; Streamlit deployment help	Bring Part 2 draft
14	M 11/23 (M25)	CSV Import + DuckDB	“ Data Import Bootcamp ”; peek under the hood (indexes, columns)	A4 due 3:30 PM
14	W 11/25	<i>No Class — Thanksgiving Recess</i>		
15	M 11/30 (M26)	CASE Statements	“ Decision Tree in SQL ”: RFM-style customer segmentation	Final Project initial submission (notebook + URL) due 3:30 PM
15	W 12/2 (M27)	Oral Assessments — Day 1	Notebook & dashboard walk-throughs	Know every query; A5 due Fri 12/4, 11:59 PM
16	M 12/7 (M28)	Oral Assessments — Day 2	Awards & wrap-up	Peer eval due 12/9; resubmission due 12/11; no final exam

Session-by-Session Guide: What We Will Do in Each Meeting

Module A — Foundations: Data as a Business Asset (Meetings 1–4)

Meeting 1 — Wed, Aug 26: What is Data? Business Runs on It. We open with “**Data in Your Pocket**”: you list ten pieces of data a business holds about you—your coffee order, your package tracking number, your streaming history—and we trace each one back to the database that stores it. Then we open Google Colab together for the first time and you type your first cells—your name, your age, a list of favorite brands. No SQL yet; the goal is comfort. *Before class*: complete the **Code of Conduct Signature** and intake survey (Canvas), and the Week 0 module if you are new to coding.

Meeting 2 — Mon, Aug 31: From Spreadsheets to Databases. “**Human Spreadsheet**”: the class arranges itself into rows and columns to feel how tables work—then we simulate what happens when two “analysts” update the same cell at once, and the case for a real database makes itself. You design your first table structure on paper, and we close with “**Organize the Chaos**”, sorting playing cards into table format.

Meeting 3 — Wed, Sep 2: Your First Database Table. You sketch a café menu table on paper and watch it become a live SQLite table; you write your first SQL statement, CREATE TABLE, and practice with a favorite-brands table. **Mini-Challenge 1 assigned**: design a table for a campus coffee-shop menu. *Before class*: Extension notebooks 1–2 (Canvas).

Meeting 4 — Tue, Sep 8: Data Types & Primary Keys (*Monday classes meet*). A table is only as trustworthy as its columns: in “**Type Casting Call**” you match real business facts—a price, a SKU, an order date, an email address—to `INTEGER`, `REAL`, and `TEXT`, and see what silently breaks when the type is wrong. We then give the café menu a proper `PRIMARY KEY` and meet the idea that every row needs an identity—the foundation for the keys and `JOINS` ahead.

Module B — Reading Data with `SELECT` (Meetings 5–9)

Meeting 5 — Wed, Sep 9: `SELECT`, Your First Real SQL. The success moment: `SELECT *` `FROM`. A “**Data Treasure Hunt**” in a provided product catalog—and by the end of class, **everyone** has run their first query.

Meeting 6 — Mon, Sep 14: Choosing Specific Columns. “**Column Detective**” on a CRM export: which columns does marketing actually need—and why pulling everything is how data leaks happen. Hands-on with `SELECT name, price FROM menu`, then partner rounds taking turns writing `SELECT` statements. *Before class:* review the string-function reference sheet.

Meeting 7 — Wed, Sep 16: `WHERE`, Finding What You Want. One condition at a time: a physical “**Filter Game**” with cards, then practice segmenting customers (`WHERE age > 18`, `WHERE price < 10`) and a live demo of the classic mistakes and what happens when `WHERE` goes wrong. **Mini-Challenge 2:** extract a specific customer segment from a customer table.

Meeting 8 — Mon, Sep 21: Multiple Conditions. `AND` and `OR`; the “**Complex Filters**” activity, where partners build a returns-abuse and fraud screen exactly the way an e-commerce risk team would. **Project kickoff:** the Data-Driven Enterprise brief and the three industry tracks are released, and team sign-up (3–4 students) opens, due **Fri, Sep 25, 11:59 PM**.

Meeting 9 — Wed, Sep 23: Simple Text Functions. `UPPER`, `LOWER`, `LENGTH`; the “**Text Transformer**” game; cleaning messy customer names and shipping addresses—and why standardization decides whether packages arrive, duplicates get merged, and marketing emails embarrass you.

Module C — Putting It Together: Challenge, Math & Data Quality (Meetings 10–13)

Meeting 10 — Mon, Sep 28: Database Puzzle Challenge. A team-based progressive puzzle: rotate through **five stations every 15 minutes**—(1) `SELECT/DISTINCT/FROM`; (2) `WHERE`, `AND/OR`, `IN`, `LIKE`, `NULL`; (3) text functions; (4) `GROUP BY/HAVING`/aggregates; (5) subqueries, set operations, `EXISTS`. **Assignment 1 due 3:30 PM.**

Meeting 11 — Wed, Sep 30: Math in SQL. Basic arithmetic in queries; “**Calculate the Bill**”: restaurant math—totals, discounts, taxes—built up step by step into the unit economics every manager reads. **Mini-Challenge 3:** a fraud-and-returns filter combining multiple conditions.

Meeting 12 — Mon, Oct 5: `NULL` and Missing Data. What `NULL` really means; `COALESCE` and `IFNULL`; “**Fill the Gaps**”; **Case Study 1:** a retail inventory with missing prices—and a short accounting of what bad data quietly costs a firm.

Meeting 13 — Wed, Oct 7: Views, CTEs, and Temporary Tables. The analyst’s report shelf: how to organize queries you will reuse—`CREATE VIEW` for the reports leadership asks for

weekly, `WITH` clauses for readable multi-step logic—directly useful for structuring your project database.

Module D — Writing & Changing Data (Meetings 14–15)

Meeting 14 — Wed, Oct 14: Adding Data with `INSERT`. `INSERT INTO`; “**Populate the Database**”—receiving day at the warehouse; partners review each other’s `INSERT` statements.

Meeting 15 — Mon, Oct 19: `UPDATE`, `DELETE`, and Transactions. Modifying existing data—and why the `WHERE` clause is a safety device. “**Fix the Mistakes**”, then the **SQL Detective Mystery**: solve a crime by correcting the data. We close with the one systems idea every businessperson should own: **transactions and ACID**—the mechanism that guarantees a bank transfer, an inventory decrement, or a double-booked seat never half-happens. **Mini-Challenge 4**: populate and correct an inventory table.

Module E — Connecting Tables: JOINS (Meetings 16–17)

Meeting 16 — Wed, Oct 21: Your First JOIN. The major concept: connecting two tables. A **physical joining demo with string**, Venn diagrams for join types, joining customers to orders on one column, and `INNER` vs `LEFT JOIN`.

Meeting 17 — Mon, Oct 26: Types of JOINS. “**Missing Data Detective**”: which join reveals customers who never purchased—and orders that never delivered—and when to use which join. **Midterm Project (Part 1) due 3:30 PM.**

Module F — Summarizing Data: The Manager’s SQL (Meetings 18–21)

Meeting 18 — Wed, Oct 28: `GROUP BY` with `HAVING` + Guest Speaker (Databricks). We open with a **guest speaker from Databricks** on how modern data teams put grouped analytics to work at warehouse scale—and what they look for when they hire. Then the “**Grouping Olympics**”: filtering groups with `HAVING` and a team competition for the best grouped query, with progressively harder practice problems.

Meeting 19 — Mon, Nov 2: Sorting and Limiting (TA leads). `ORDER BY` and `LIMIT`; the “**Top 10 Challenge**”; a data-analysis competition to build best-seller and top-customer leaderboards from sales data. **Mini-Challenge 5**: two-store analysis with `JOIN`.

Meeting 20 — Wed, Nov 4: Aggregate Functions (TA leads). `COUNT`, `SUM`, `AVG`, `MIN`, `MAX`—the five verbs behind every KPI dashboard; “**Store Statistics**”; seeing aggregates with physical objects before writing them in SQL.

Meeting 21 — Mon, Nov 9: `GROUP BY` in Depth. Group **before** you aggregate; “**Group and Count**” with classroom items; a visual demo of how grouping actually works—then cohorts and segments, the grouped queries behind retention curves. **Assignment 2 due 3:30 PM.**

Module G — Advanced Querying (Meetings 22–23)

Meeting 22 — Wed, Nov 11: Three-Table JOINS. “**Data Web**”: connect customer → order → product by adding one table at a time—and the ambiguous-column pitfall that trips everyone once. **Mini-Challenge 6**: customer–order analysis with `GROUP BY`.

Meeting 23 — Mon, Nov 16: Subqueries. A query within a query; “**Nested Questions**”; finding above-average values—which sellers beat their category’s average delivery time, which stores beat the chain—a pattern you will reuse directly in the project. **Assignment 3 due 3:30 PM.**

Module H — Project Sprint & The Modern Analytics Stack (Meetings 24–26)

Meeting 24 — Wed, Nov 18: Project Workshop Day. A full session of team work with instructor and TA support: a **milestone check** on your analysis plan and draft multi-table queries, plus a **live Streamlit deployment help station**. **Bring your Part 2 draft notebook.**

Meeting 25 — Mon, Nov 23: Importing CSV Files + DuckDB. “**Data Import Bootcamp**”: pull a real CSV from one of the curated sources—or a slice of the NYC taxi data—and query it at warehouse grade with **DuckDB**, inside Colab, zero installation—plus troubleshooting the common import errors. **Assignment 4 due 3:30 PM.**

Meeting 26 — Mon, Nov 30: CASE Statements. IF–THEN logic in SQL; “**Decision Tree in SQL**”; RFM-style customer segmentation—label your customers new, loyal, or at-risk in one query—the last technique for your project. **Final Project initial submission (notebook + live URL) due 3:30 PM.**

Module I — Showcase & Oral Assessments (Meetings 27–28)

Meeting 27 — Wed, Dec 2: Project Walkthroughs & Oral Assessments, Day 1. Assigned groups (by group number, announced on Canvas) walk the class through their notebooks and **live dashboards**; each member **answers questions about the team’s queries**; live feedback and scoring. **Assignment 5 (The SQL Gauntlet) due Fri, Dec 4, 11:59 PM.**

Meeting 28 — Mon, Dec 7: Oral Assessments, Day 2 & Celebration. Remaining groups present; awards (Best Project, Most Creative, Best Analysis) and peer nominations; course wrap-up and celebration. **Peer evaluation due Wed, Dec 9; final resubmission due Fri, Dec 11. No final exam.**

Delivery Method

For almost all modules, some material is best delivered asynchronously (through videos I record for you)—basically lecture material—while more interactive material is best delivered synchronously in our in-person class meetings. Each module is therefore divided as follows:

Before Class (15–20 minutes)

- Micro-learning videos (5–7 minutes each) covering core concepts
- Pre-class Colab notebook with guided examples to run through
- Quick knowledge check (2–3 **Mentimeter** questions)
- Optional: the week’s SQLBolt lesson and CS50 SQL deep-dive video

You can post any questions or comments in the “**Q & A Board**” **discussion board** on Canvas. Before posting, be sure your question hasn’t already been asked (and possibly answered); if you have an answer to another student’s question, please provide it—I strongly encourage you to help your fellow students. Your TA and I monitor the boards periodically.

In-Person Class Session

Opening (10 minutes): quick review of pre-class concepts via [Kahoot!](#); questions from the Q&A board.

Interactive Coding Segments (50 minutes):

- **Live Coding (15 min):** instructor demonstrates new concepts in Colab
- **Pair Programming (15 min):** students work in pairs on guided exercises
- **Mini-Challenge (10 min):** individual quick problem-solving
- **Share & Debug (10 min):** students share solutions, troubleshoot together

Wrap-up (15 minutes): reflection (what worked / what was challenging); preview of the next class; **exit ticket** (one thing learned + one question).

After Class

- Practice Colab notebook with progressive difficulty
- Peer review activity (students review each other's code)
- Optional challenge problems for advanced students (Enrichment module)

Class Attendance

You are expected to attend all classes and to participate regularly in class discussions. Absences due to documented medical or family emergencies and major religious holidays will be excused (**at most, two excused absences will be granted**). In addition to the two documented/excused absences, you may attend up to two additional classes via Zoom, which will be counted as present. For each unexcused absence/leave and/or **negative behavior**, you will receive **negative points** as your grade for that day, and your overall in-class participation grade will be reduced. You are responsible for anything covered in class on the days you miss, and your assignments are due on the specified dates and times regardless of whether you attend class.

Please bring your **name tent** to **every class meeting**—it helps your TA assign participation points and enhances class discussions.

Remember: everyone was a beginner once. Be patient with yourself, help your classmates, and enjoy discovering the power of data! **Let's start this journey together!**

Instructor & Support

Instructor:	Emisa Nategh (About Dr. Nategh page on Canvas)
Email:	en471@georgetown.edu
Office:	Hariri Building, Floor 5-502
Office Hours:	By appointment: https://calendar.app.google/ePjfmbsbcGBEB5kp8 . Between appointments, your 24/7 Gemini course tutor is always on.

Communication

I will communicate with you using **Canvas Announcements**; please **ensure that your Canvas notifications are set accordingly**. **Post all non-confidential, course-related questions on the “Q & A Board” first**—others benefit from the answers, and I usually respond within 24 hours to discussion posts; my response to email may be slower. Email remains the most efficient method for personal matters, or to set up a time to meet in person or over Google Meet/Zoom.

Support: Who should you contact?

- **Your 24/7 AI Course Tutor — the OPAN 3257 SQL Tutor Gem** (custom Gemini; link on Canvas; free with your Georgetown Google account and the no-cost AI Pro student plan): the **first stop** for concept questions, practice problems, and debugging help at any hour—an always-open tutor, teaching assistant, and virtual office hour.
- **Your Instructor, Dr. Nategh** (en471@georgetown.edu): questions regarding course content/curriculum, and anything the AI tutor cannot settle.
- **Your TA & Grader: TBD**: questions regarding course materials, assignments, quizzes, and the group project. TA office hours (twice weekly; Zoom link under “Zoom Conferencing”) posted on Canvas.

Zoom Best Practices

Zoom is used for office hours and recorded materials, and is our backup platform for live sessions (links on Canvas):

- Join 10 minutes early to resolve any tech issues and show respect for others’ time.
- Keep your camera ON to stay engaged; if you have camera issues, inform me.
- Use nonverbal cues (thumbs up/down); display your full name and headshot.
- Use the chat for questions and feedback, but wait for chat breaks to send messages.
- Mute your microphone when not speaking; hold the <SPACE> bar for temporary unmute.
- Avoid distracting backgrounds; turn off email notifications and your cell phone.

Mentimeter/Kahoot!

We will use Mentimeter to make class more engaging, understand your knowledge, and provide feedback. **Mentimeter/Kahoot! questions count toward participation**; some will have correct answers, others will not. You need a device—smartphone, tablet, or laptop—to scan the barcode or use the link. Mentimeter/Kahoot! use must adhere to Community Standards: do not look at others’ devices or use multiple remotes or accounts. Remote participation is allowed if needed. Mentimeter aims to enhance engagement, not just quiz or take attendance.

Generative AI

Generative AI tools can assist with idea generation, analysis, and writing, and act as tutors for complex concepts—and this course **provides** you with a course-customized Gemini Gem (the *OPAN 3257 SQL Tutor*, 24/7, free on your Georgetown account with the no-cost AI Pro student plan) plus Colab’s built-in Gemini assistant, precisely so you can get unstuck while practicing and debugging. However, don’t rely on them blindly: **AI tools confidently lie and may fabricate information. Always critically assess their output, as you will be held accountable for any errors.**

Some companies restrict AI use due to cyber risks and regulations, so you must be prepared to manage without AI—and to perform **better** than AI tools when they are available. A metaphor: there are robots that can move fast and lift heavy weights, but if you need to become physically fit, you will not have robots do your workouts for you in the gym. Learning is similar—**if you are in school to learn, you have to do the work yourself.**

Do not upload copyrighted course materials or assessment content into generative AI tools—doing so violates copyright law and academic integrity; similarly, do not upload or download copyrighted materials from the internet. If you use Gemini, ChatGPT, Claude, or similar tools, **credit them in your work as you would a team member** and cite any direct language used.

Generative AI tools are NOT allowed on the Semester Project (Parts 1 and 2); tools like Grammarly for language aid are permitted. My assessment philosophy: **welcome the tools, verify the understanding**—credit is earned through work that cannot be delegated, which is why the final project is defended orally in class. We will discuss how to use AI tools effectively for data work and for learning.

Canvas Submission Rules

All submissions must be made through Canvas. When submitting a file, ensure that you have both **uploaded and submitted** it, and verify that Canvas confirms your assignment as submitted; if I do not receive your submission, it will be considered late and subject to penalties. Where resubmission is allowed (e.g., the final project resubmission), **only the most recent submission will be graded** unless otherwise instructed—make sure your latest upload is complete, including all files, answers, and the dashboard URL. **Avoid special symbols in file names** (e.g., #, \$, %, @, &, *, /, quotation marks, ;, :, .,), as these may prevent me from accessing your files, resulting in late penalties. For team submissions, include your team number, submit only **one** file per team, and list all team members’ names in the first notebook cell; late team submissions incur penalties for the entire team, regardless of individual responsibility.

Late Submission Policy

1. Late submissions lose **20% per hour**.
2. Canvas rounds up to the nearest hour for penalty calculation.
3. Example: 1 hour 15 minutes late = Canvas 2-hour penalty = 40% deduction.
4. **No points are awarded after solutions are posted or discussed in class.**

Timely submissions are crucial for course operations.

Regrades and Makeups

If you believe there is a grading error, please email the entire assignment with a written explanation **within one week** of the grade being posted. Note that **the entire assignment will be regraded, and your grade may go up or down**. Makeups for missed in-class participation and mini-challenges are not available, regardless of the reason; however, your **lowest mini-challenge is dropped**, and participation points accumulate from many activities, so you can make up for missed points by contributing more to other activities.

Course Software and Textbook Information

No textbook is required. Many students find the interactive Colab notebooks, videos, and class notes sufficient for studying.

Recommended: Ramakrishnan, R., Gehrke, J., *Database Management Systems*, McGraw-Hill. Free companions: the CS50 SQL lecture notes (<https://cs50.harvard.edu/sql/>), and the official PostgreSQL tutorial (<https://www.postgresql.org/docs/current/tutorial.html>).

Software: **Google Colab** (SQLite and DuckDB) — free, cloud-based, browser-only; access <https://colab.research.google.com> with your Georgetown Google account and you're ready to start coding. For the final-project deployment, free **GitHub** and **Streamlit Community Cloud** accounts (guide on Canvas). **No installation, ever.**

Industry Connection: Guest Speaker & Site Visit

- **Practitioner guest speaker:** at least one guest speaker from the data industry visits this course during the semester (company and date announced on Canvas; recent course guests have come from firms such as MongoDB, Databricks, IBM, and Deloitte). The session is **recorded and shared** so all students benefit.
- **Site visit (optional):** an optional visit to the **QTS Data Centers campus**—see where databases physically live—will be organized at **no cost to students**. Date, transportation, and sign-up on Canvas; capacity may be limited.

Student Voice: Feedback That Changes the Course

I field my own **anonymous feedback survey twice each semester**—once around week 4 and again around week 8—and then I visibly change the course in response, so you can watch a week-4 comment reshape the week-5 plan. Every session includes rapid, low-stakes checks (Mentimeter/Kahoot!), and several classes close with a short exit-ticket quiz. Structured peer review through **FeedbackFruits** keeps collaborative workload equitable and every member's contribution visible.

Tech Support

Support Within Canvas

Students have 24/7 access to Canvas technical support, which includes live chat and a support hotline at **855-338-2770**. Clicking the “Help” icon in the lower-left of your Canvas window shows all available support and feedback options. For help on a specific feature, check out the Canvas Student Guide.

Support Within Georgetown

Contact the University Information Services (UIS) Center at Help@georgetown.edu or 202-687-4949 if you have a question regarding: your GU NetID and password; your GU email account; any connectivity issues.

Additional Information

University Guidelines and Academic Policies: see the *University Guidelines and Academic Policies* file posted on Canvas (course syllabus module). All students are bound by the Georgetown University Honor System (<https://honorcouncil.georgetown.edu/system/policies>); assignments may be submitted to Turnitin for plagiarism detection.